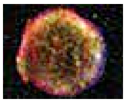
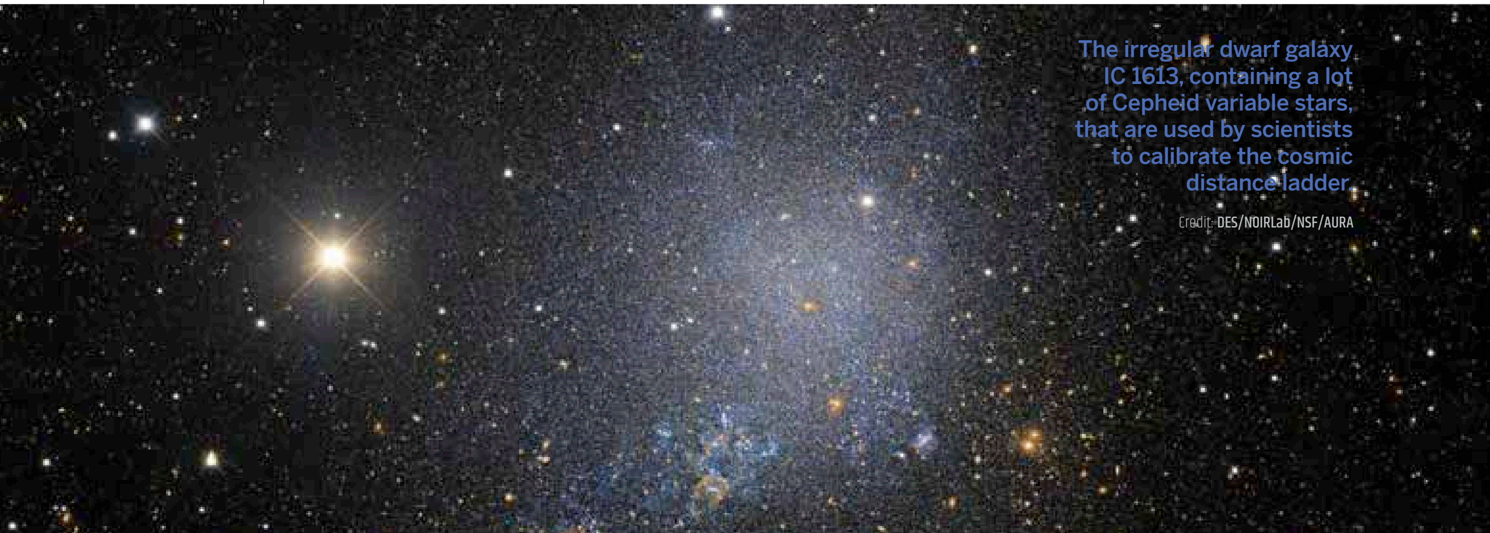


Uranium crystals trigger supernova
White dwarves could go supernova in response to nuclear fission caused by Uranium crystals.
http://dgit.in/apr21-51

Starlink rocket misses landing
SpaceX's latest Starship rocket prototype, SN11 crashed during its landing attempt.
http://dgit.in/apr21-52



The irregular dwarf galaxy IC 1613, containing a lot of Cepheid variable stars, that are used by scientists to calibrate the cosmic distance ladder.

Credit: DES/NOIRLab/NSF/AURA

DARK ENERGY DATA DUMP

A massive catalogue of the most distant objects has been released to the public

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The Dark Energy Survey (DES) was a massive collaboration between a number of scientists and institutions around the world, that looked at the most distant objects in the night sky for a period of over six years, to study dark matter and dark energy throughout the universe. Over 500 scientists from America, South America and Europe were a part of the collaboration, and began in 2013. The Data Release 2 is the second dark energy data dump by the DES project, and took about six years to capture. During the observation campaign, one eighth of the sky or a 5000 degree square area was surveyed, imaging stars, galaxies, quasars, interstellar gas clouds, supernovae and other objects. If you think that sentence is a mouthful, try saying the next one out loud in one breath. It was also used to detect trans Neptunian objects, 16 new Jovian moons, asteroids, captured the largest superluminous supernova explosion ever detected, recorded the most distant supernova event, discovered new distant dwarf planets, found additional evidence of dark matter halos, detected the collision between two collapsed stars (also called a kilonova), found a number of dwarf satellite galaxies in orbit around the Milky Way, and also detected a collision between two neutron stars in the visible wavelengths. That last one was an event that was also picked up by the LIGO and VIRGO gravity wave detectors. The DES was very much a part of the new era of multimessenger astronomy.

The imaging instrument was a fourteen foot long specially built camera known as the Dark Energy Camera, with an imaging resolution of 570 megapixels. The light is captured on an array of 62 2048x4096 specially built state of the art CCDs. The light from the most distant objects are extremely redshifted, and it is easier to detect them if the silicon chips used for the CCD sensors are thicker. For this reason, the CCD sensors in the Dark Energy Camera are ten times thicker than the CCDs used in a conventional consumer camera. There are additional "guider" CCDs as CCDs for focus and alignment. The camera operates at low temperatures, -100 degrees Celsius to reduce noise, and is cooled by liquid nitrogen. To prevent the optical components from fogging up, the camera is also operated in vacuum.

The survey measured the precise characteristics of the light, including its redshift, that allows astronomers to precisely map the